

Challenge (Habanero)

- Q1.** Factor the expression: $x^2 + 5x + 4x + 20$
(A) $x(x + 5) + 4(x + 5)$
(B) $x(x + 4) + 5(x + 4)$
(C) $(x + 4)(x + 5)$
(D) $x(x + 3) + 4(x + 5)$
(E) $(x + 2)(x + 10)$
- Q2.** Factor: $3x^2 + 12x + 4x + 16$
(A) $3x(x + 4) + 4(x + 4)$
(B) $3x(x + 2) + 8(x + 2)$
(C) $3(x + 4)(x + 2)$
(D) $(3x + 4)(x + 4)$
(E) $3(x + 2)(x + 4)$
- Q3.** A construction worker needs to build a fence with an area of $x^2 + 7x + 12x + 84$. Factor the area to find the length and width.
(A) $(x + 7)(x + 12)$
(B) $(x + 3)(x + 28)$
(C) $(x + 4)(x + 21)$
(D) $(x + 6)(x + 14)$
(E) $x(x + 7) + 12(x + 7)$
- Q4.** Simplify the expression: $2x^2 + 10x + 5x + 25$
(A) $2x(x + 5) + 5(x + 5)$
(B) $2(x + 5)(x + 2)$
(C) $(2x + 5)(x + 5)$
(D) $2x(x + 2) + 5(x + 5)$
(E) $2(x + 2)(x + 5)$
- Q5.** A chef is making a recipe with an area of $x^2 + 9x + 6x + 54$. Factor the area to find the length and width.
(A) $(x + 9)(x + 6)$
(B) $(x + 3)(x + 18)$
(C) $(x + 6)(x + 9)$
(D) $x(x + 6) + 9(x + 6)$
(E) $x(x + 9) + 6(x + 9)$
- Q6.** Factor: $2x^2 + 8x + 3x + 12$
(A) $2(x + 2)(x + 3)$
(B) $2x(x + 4) + 3(x + 4)$
(C) $2(x + 3)(x + 2)$
(D) $2x(x + 2) + 4(x + 3)$
(E) $x(2x + 3) + 4(2x + 3)$
- Q7.** A builder needs to create a foundation with an area of $x^2 + 8x + 15x + 120$. Factor the area to find the length and width.
(A) $(x + 8)(x + 15)$
(B) $x(x + 8) + 15(x + 8)$
(C) $(x + 5)(x + 24)$
(D) $(x + 10)(x + 12)$
(E) $x(x + 15) + 8(x + 15)$
- Q8.** Factor the expression: $x^2 + 6x + 5x + 30$
(A) $x(x + 6) + 5(x + 6)$
(B) $(x + 5)(x + 6)$
(C) $x(x + 5) + 6(x + 5)$
(D) $x(x + 3) + 5(x + 6)$
(E) $x(x + 6) + 5(x + 5)$

Open Ended Questions (FRQ)

- Q9.** Factor the expression: $3x^2 + 14x + 5x + 20$. Verify your answer by multiplying.
- Q10.** A chef is making a new recipe with an area of $2x^2 + 11x + 7x + 35$. Factor the area to find the length and width. Verify your answer by multiplying.

Chef's Tips

- Check for common factors before grouping
- Use the distributive property to verify answers
- Apply factoring by grouping to real-world problems